East West University

Lab Report

Semester: Summer-2025

Course Title: Electronic Circuits **Course Code:** CSE251

Section: 04

Expt No: 03
Expt Name: Study of Zener Diode
Group No: 06
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Title: Study of Zener Diode

Objectives:

1. To measure the I-V (Current vs. Voltage) characteristics of a Zener diode.

2. To check how well the Zener diode keeps the voltage steady when the load or supply voltage changes.

Theory (Summary):

A Zener diode is a special type of diode. It is made to work in the reverse breakdown region, where normal diodes can get damaged. The Zener diode is used in voltage regulator circuits because it can keep the voltage almost the same even if the input voltage or the load changes.
When the reverse voltage becomes equal to or more than the Zener breakdown voltage, the diode starts conducting a lot of current but the voltage across it stays almost the same. This helps in giving a stable output voltage.

There are three main regions of Zener diode operation:

1. Forward region: Works like a normal diode.

2. Leakage region: Small current flows before breakdown.

3. Breakdown region: Voltage stays steady, current increases.

The Zener diode helps in making power supplies and circuits safe and stable by keeping the voltage fixed.

Equipment:

1. Zener diode (6.2V)
2. Resistance (220 Ω, 470 Ω, 1 kΩ)
3. POT 10 kΩ
4. DC Power supply
5. Breadboard
6. Multimeter
7. Ammeters

Circuit Diagram:

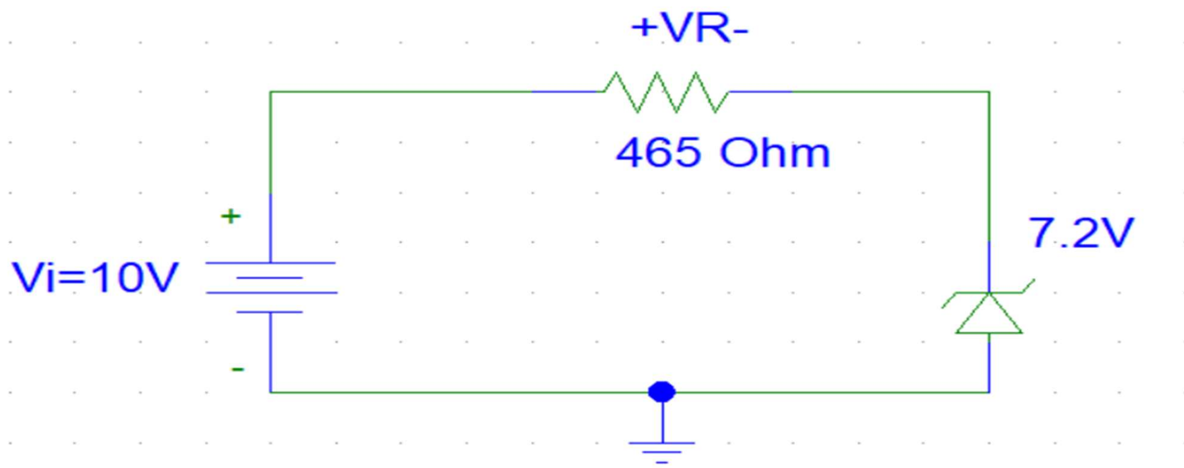


Figure-1: Circuit with no load

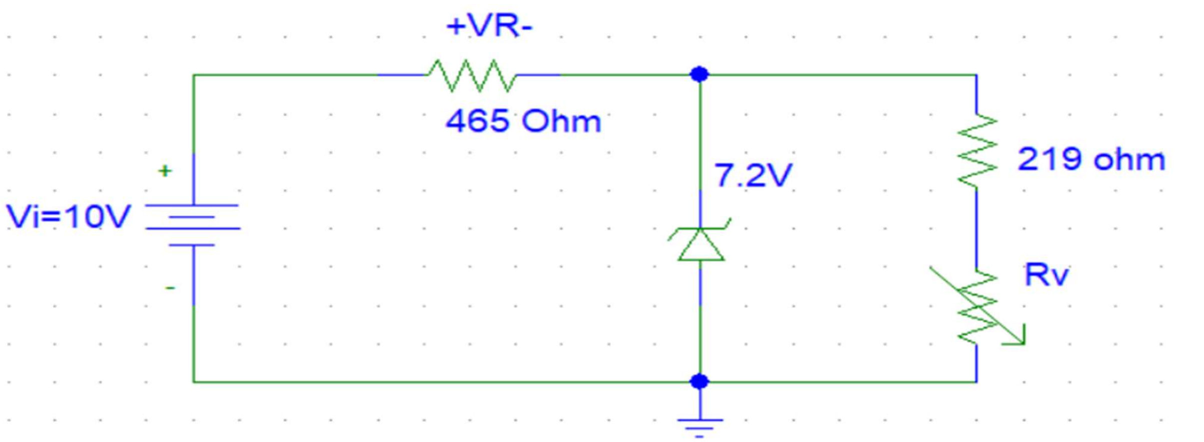


Figure-2: Circuit with variable load

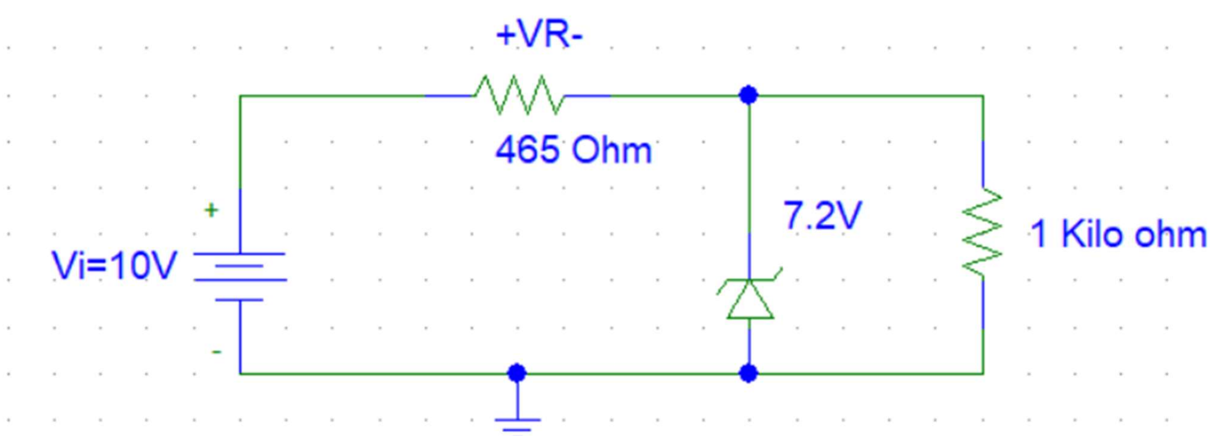


Figure-3: Circuit with variable Supply voltage

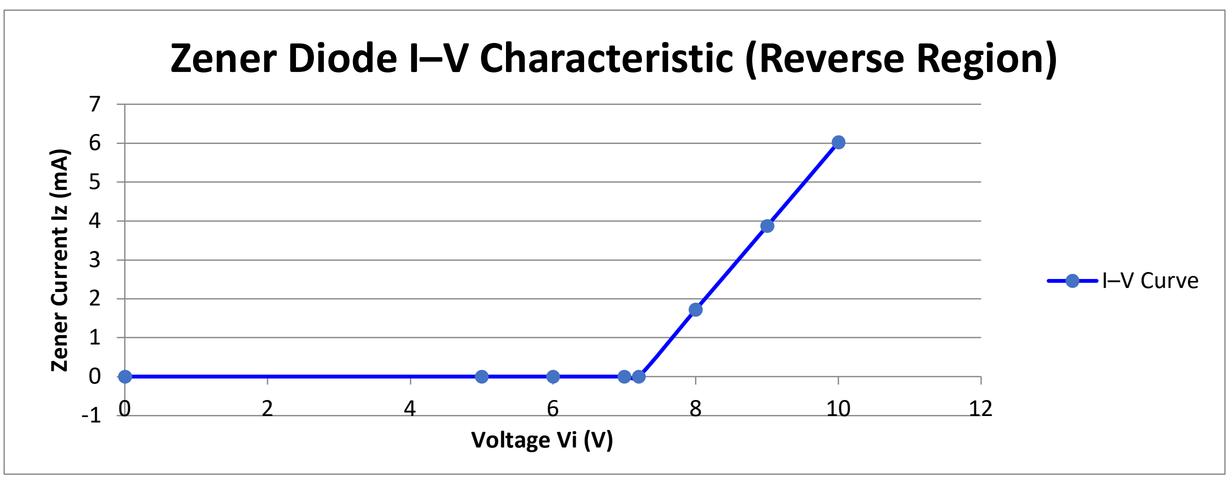
Pre-Lab Report:

1. Draw the I–V characteristic of Fig. 3 (circuit with no load) using experimental values.

Answer

Since there's no load (open circuit at the output), the only current flows through the series resistor and into the Zener diode. We can vary v_i and measure v_z (which remains constant after breakdown) and current I_z. Calculate theoretical values for the I-V plot (idealized):

v_i (V)	v_z (V)	$v_r = v_i - v_z$ (V)	$I = v_r / R$ (mA)
0	0	0	0
5	5	0	0
6	6	0	0
7	7	0	0
7.2	7.2	0	0
8	7.2	0.8	1.72
9	7.2	1.8	3.87
10	7.2	2.8	6.02



Zener Diode Behavior:

1. For $v_i < 7.2V$, the diode doesn't conduct (I = 0).

2. For $v_i \geq 7.2V$, it enters breakdown and maintains $v_z = 7.2V$.

2. Measure V_L, V_R, I_z for 10 V supply voltage.

Answer

From the circuit,

$$V_L = V_Z = 7.2\text{ V}$$

$$V_R = V_i - V_Z = 10 - 7.2 = 2.8\text{ V}$$

$$I_Z = \frac{V_R}{R} = \frac{2.82.8\text{ V}}{465\ \Omega} = 0.00602\text{ A} = 6.02\text{ mA}$$

3. For $R_L = 220\Omega$, verify the condition for conduction.

Answer

Now, the Zener is in parallel with RLR_LRL, so total current from the source is split:

Let:

I = total current through 465 Ω

$$I_L = \frac{V_Z}{R_L} = \frac{7.2}{220} = 32.73\text{mA}$$

$$I = \frac{(10 - 7.2)}{465} = 6.02\text{mA}$$

$$I_Z = I - I_L = 6.03\ mA - 32.73\ mA = -26.71\ mA.$$

So total current from source is only 6.02 mA, but the load alone requires 32.73 mA. This is a contradiction!

Zener will not conduct (no regulation) because the supply current is insufficient to drive both load and Zener. The voltage will drop below 7.2 V, violating the Zener conduction condition.

So, Zener diode will conduct current if $I_Z = I - I_Z > 0$.

4. Measure $R_{L_{min}}$, $R_{L_{max}}$, and $I_{L_{min}}$ for c = 7.2 mA and 10 V supply voltage in Fig. 4.

Answer

Given,

Input Voltage, Vi =10V

Zener Diode Voltage, Vz=7.2 V

Series Resistor, Rs=465 Ω

Load Resistance R_L, =219 Ω

Maximum Load Current $R_{L_{min}}$=7.2V

Calculate the total current (I_T) through the series resistor:

$$V_R = V_i - V_Z = 10\text{ V} - 7.2\text{ V} = 2.8\text{ V}\ I = \frac{V}{R} = \frac{2.8\text{ V}}{465\,\Omega} \approx 0.0060215\text{ A} = 6.02\text{ mA}$$

Calculate $I_{L_{min}}$:

We know $I_T = I_Z + I_L$

we use $I_Z = I_{Zmax}$

$$I_{L_{min}} = I_T - I_{Z_{max}} = 6.02\text{ mA} - 7.2\text{ mA} = -1.18\text{ mA}$$

Calculate RL_{min} :

$$R_{L_{min}} = \frac{I_{L_{min}}}{V_Z} = \frac{-1.18\times10^{-3}\text{ A}}{7.2\text{ V}} \approx -6101.69\,\Omega$$

Calculate R_{Lmax} (considering the physical minimum load current):

The actual minimum load current $I_{L_{min\_actual}}$ is 0 A (open circuit)

$$R_{L_{max}} = \frac{V_Z}{I_{L_{min\_actual}}} = \frac{7.2\text{ V}}{0\text{ A}} = \infty$$

5. Measure $V_{i_{min}}$ and $V_{i_{max}}$ for $I_{Z_{max}}$ = 7.2 mA of Fig.5

Answer

Given,

Zener Voltage, Vz=7.2 V

Series Resistor, Rs=465 Ω

Load resistor $R_L = 1\text{ k}\Omega = 1000\ \Omega$

$I_{Z_{max}} = 7.2$ mA

V_i is the input voltage (we are to find min and max values)

Find Load Current I_L

Since the load is connected in parallel with the Zener diode,

it shares the same voltage $V_Z = 7.2$ V.

$$I_L = \frac{V_Z}{R_L} = \frac{7.2}{1000} = 7.2\text{ mA}$$

Total Current from Source I_R (when $I_Z = I_{Z,\max} = 7.2$ mA)

$$I_R = I_Z + I_L = 7.2\text{ mA} + 7.2\text{ mA} = 14.4\text{ mA}$$

Ohm's Law to Find $V_{i_{max}}$

$V_i = V_Z + (I_R \times R) = 7.2\text{ V} + (14.4\text{ mA} \times 465\ \Omega) = 7.2 + 6.696 = \boxed{13.896\text{ V}} \approx \boxed{13.9\text{ V}}$

This is when all the current goes to the load, and no current flows through the Zener diode:

$I_R = I_L = 7.2$ mA

$V_i = \boldsymbol{V_Z} + (\boldsymbol{I_R} \times \boldsymbol{R}) = 7.2 + (7.2\text{ mA} \times 465) = 7.2 + 3.348 = \boxed{10.548\text{ V}} \approx \boxed{10.55\text{ V}}$

Final Answers:

Minimum Input Voltage $V_{i_{min}} \approx 10.55$ V

Maximum Input Voltage $V_{i_{max}} \approx 13.9$ V

These values ensure proper Zener regulation, where $I_Z \leq 7.2$ mA

Experimental Datasheet:

Table 1: Data for I-V Characteristics

Vi (Volt)	**V_R (volt)**	**V_Z (volt)**	**$I_Z = \frac{V_R}{R}$ (mA)**
1	0	0.9	0
2	0	1.9	0
3	0.1	2.7	0.215
4	0.6	3.3	1.29
5	1.3	3.6	2.80
6	2.1	3.8	4.52
7	2.9	3.9	6.24
8	3.8	4	8.17
9	4.7	4.1	10.11
10	5.6	4.1	12.04

Table 2: Data for Regulation Due to Load Variation

Vi (Volt) =10 V	V_L (volt)	I_L (Amp)
10	3.6	0.4 mA (10.4 K Ω)
		0.2 mA (9.08 K Ω)
		0.19 mA (8.01K Ω)
		0.17 mA (7.05 K Ω)
		0.17 mA (6.03 K Ω)
		0.16 mA (5.08 K Ω)
		0.14 mA (4.03 K Ω)
		0.11 mA (3.01 K Ω)
		0.11 mA (2.04 K Ω)
		0.09 mA (1.07 K Ω)

Table 3: Data for Regulation Due to Supply Voltage Variation

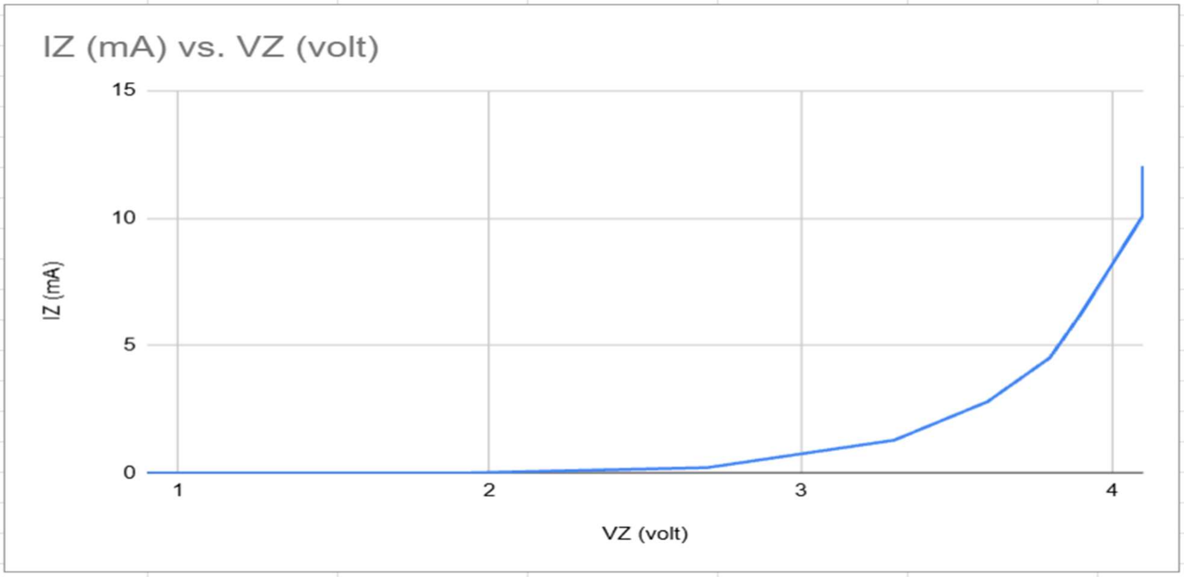
V_i (Volt)	V_L (Volt)
1	0.63
2	1.28
3	1.99
4	2.57
5	3.09
6	3.41
7	3.65
8	3.82
9	3.95
10	4.05

Post Lab Report Questions:

1. Plot the I-V characteristics of the Zener diode. Determine the Zener breakdown voltage from the plot.

Answer

VZ (volt)	IZ (mA)
0.9	0
1.9	0
2.7	0.215
3.3	1.29
3.6	2.8
3.8	4.52
3.9	6.24
4	8.17
4.1	10.11
4.1	12.04



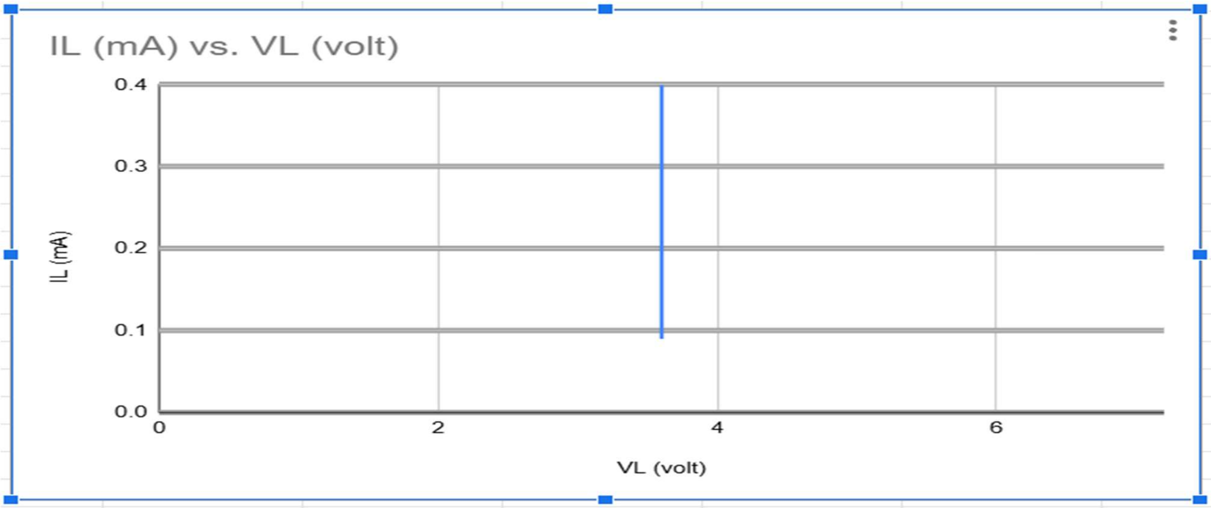
At Vz = 0.9V to 3.9V, the current Iz increases slowly. But when Vz reaches 4.0V to 4.1V, Iz suddenly jumps from 8.17 mA to 10.11 mA. Even though Vz stays constant at 4.1V, the current keeps rising (to 12.04 mA).

This means the Zener diode enters full breakdown at around 4.1V.

2. Plot I_L Vs V_L for the data of table-2. Find the load regulation and compare it with the pre-lab data.

Answer

V_L (volt)	I_L (mA)
3.6	0.4
3.6	0.2
3.6	0.19
3.6	0.17
3.6	0.17
3.6	0.16
3.6	0.14
3.6	0.11
3.6	0.11
3.6	0.09



Calculation of Load Regulation from Experimental Data (Table 2):

Load Regulation (LR) quantifies how well a voltage regulator maintains a constant output voltage (V_Z) despite changes in load current (I_L). It is calculated using the formula:

$$\text{Load Regulation (\%)} = \frac{V_{\text{NL}} - V_{\text{FL}}}{V_{\text{FL}}} \times 100\,\%$$

Where:

V_{NL} = No-Load Voltage (output voltage at minimum load current)

V_{FL} = Full-Load Voltage (output voltage at maximum load current)

Substituting these values into the formula:

$$\text{Load Regulation (\%)}=\frac{(3.6\text{ V}-3.6\text{V})}{3.6\text{ V}} \times 100\,\%$$

$$\text{Load Regulation (\%)} = \frac{0\text{V}}{3.6\ V} \times 100\,\%$$

$$\text{Load Regulation (\%)} = 0\,\%$$

Comparison with Pre-Lab Data:

The pre-lab analysis (specifically from pre-lab answer 2) determined the theoretical output voltage (V_Z) for a 10V supply voltage to be **7.2 V**, based on the Zener diode's specified voltage in the circuit diagram (Fig. 3).

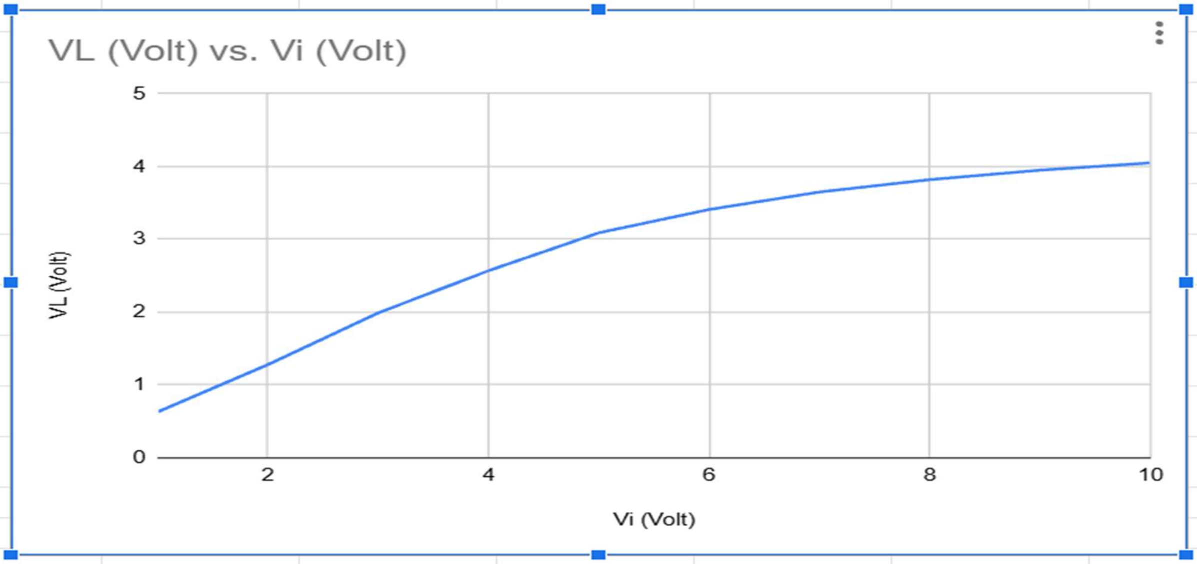
Comparison:

Pre-Lab Expectation: The pre-lab predicted a regulated output voltage of 7.2 V.

Experimental Result (Table 2): The experimental data consistently shows a regulated output voltage of 3.6 V.

3. Plot VL Vs V for the data of table 3. Find the line regulation compare it with the pre-lab data.

Answer



$$\text{Line Regulation} = \frac{\Delta V_L}{\Delta V_i} \times 100\%$$

Using the data:

$V_{L_{\min}} = 0.63\,V$ at $V_i = 1\,V$

$V_{L_{\max}} = 4.05\,V$ at $V_i = 10\,V$

So, $\Delta V_L = 4.05 - 0.63 = 3.42\,V,$

$\Delta V_i = 10 - 1 = 9\,V$

$\text{Line Regulation} = \frac{3.42}{9} \times 100\% \approx 38.0\%$

Comparison with Pre-Lab Expectation:

In the pre-lab,

Ideal Zener voltage = 7.2 V

Once breakdown starts (Vi > 7.2 V), VL remains constant.

But in the experimental result, even at Vi = 10 V, VL = 4.05 V (much lower than 7.2 V).

Discussion:

This experiment aimed to explore the behavior of a zener diode, speaially its I-V characteristies and its ability to regulate voltage when either the load or the input ~~va~~voltage changes.

From the I-V characteristic plot I_Z vs V_Z, the zener diode starts to conduct significantly at around 4.1V. This is notably different from the theoretical zener voltage of 7.2V mentioned in the pre-lab. This discripancy might be due to the specific zener diode used in the experiment or slight variations in computant tolerances.

When load regulation was examined, ~~the~~ Table-2 and I_L vs V_L graph showed a consistent output voltage (V_L) of 3.6V, regardless of the changes in load current, resulted in a load rgulation of 0%, means excellent load regulation. However, it is lower than the expected 7.2V from the pre-lab analysis.

For line regulation, Table-3 and V_L vs V_i graph showed a line regulation of 38%, means V_L changes as V_i ~~varri~~ varied. ~~Heo~~ However V_L never reached 7.2V, even with a 10V input, only reaching 4.05V

In conclusion, while the zener diode demonstarated good load ~~re~~ regulation, but also deviated significantly, suggests that the zener diode used likely had a lower breakdown voltage (closer to 3.6V or 4.1V) than the spacified 7.2 V.